

GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.1.7

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LS 1.1.7- DRUG METABOLISM

DRUG METABOLISM

LS 1.1.7: LEARNING OBJECTIVES:

- Define the term “Metabolism”
- Describe the processes of drug metabolism
- Describe the processes of Phase I and Phase II metabolism

DRUG METABOLISM

INTRODUCTION:

Metabolism is the general term for biotransformation

Biotransformation: the process by which the body changes the chemical structure of a drug to another form called a metabolite

The liver is the principal site of drug metabolism

Other sites for drug metabolism are the kidney, gut

DRUG METABOLISM

INTRODUCTION ... CONT'D:

In metabolism, drugs are changed in two major ways:

1. Reduction in lipid solubility - the drug molecule is made more water soluble and therefore easier to eliminate in the urine
2. Alteration in biological activity - conversion of a pharmacologically active substance to one that is inactive or to another active substance and

DRUG METABOLISM

Drug metabolism occurs in two broad phases:

1. Phase I metabolism: Involves chemical alteration of the basic structure of the drug e.g. oxidation, reduction or hydrolysis
2. Phase II metabolism: Involves conjugation e.g. sulphation, glucuronidation, methylation or acetylation

PHASE I REACTIONS

- Consist of oxidation, reduction or hydrolysis
- The products are more often chemically reactive, sometimes pharmacologically active, toxic or carcinogenic
- A relatively reactive group (e.g. hydroxyl) is introduced into the molecule (functionalization)
- The functional group then serves as the point for conjugation e.g. glucuronidation, in a phase II reaction

PHASE I REACTIONS: OXIDATION & REDUCTION REACTIONS

- The enzymes involved in most oxidation and reduction reactions are cytochrome P450 enzymes (also referred to as hepatic microsomal enzymes or mixed function oxidases). They are haem proteins and they have redox properties.
- The cytochrome P450 system comprises large families of related but distinct enzymes
- Oxidation reactions are more common than reduction reactions.

CYTOCHROME P-450 ENZYMES

- The cytochrome P450 families are referred to using an Arabic numeral (e.g. CYP1, CYP2 etc.). Each family has a number of sub-families denoted by an upper case letter (e.g. CYP2A, CYP3A etc.). The individual enzymes within each sub-family are denoted by another Arabic numeral (e.g. CYP2C9, CYP3A4 etc.)
- The most important cytochrome P450 enzymes in human metabolism are CYP1A2, CYP2C9, CYP2C19, CYP2D6 and CYP3A4

- CYP3A sub-family is responsible for up to half of the total cytochrome P-450 in the liver and accounts for about 50% of the metabolism of clinically important drugs. CYP 3A4 is a particularly abundant enzyme.

CYTOCHROME P-450 ENZYMES GENETIC POLYMORPHISM

- Cytochrome P-450 enzymes exhibit genetic polymorphism i.e. there are genetically determined differences in their activity of the among individuals, including differences among racial and ethnic groups
- This is a source of variation in drug metabolism in humans

CYTOCHROME P-450 ENZYME INHIBITION AND INDUCTION

- A number of factors inhibit or induce cytochrome P450 enzymes
- Many clinically important interactions between drugs are due to inhibition or induction of P450 enzymes
- Inhibition can be competitive or non-competitive and results in reduced metabolism of drugs and endogenous substrates that are metabolized by the inhibited enzymes
- Inhibitors include: ethanol (acute intake), ketoconazole, allopurinol, disulfiram, cimetidine

CYTOCHROME P-450 ENZYME INHIBITION AND INDUCTION ... CONT'D

- Induction means increase synthesis of the enzymes and results in increased metabolism of drugs and endogenous substrates metabolized by the induced enzymes
- Inducers include: rifampicin, carbamazepine, barbiturates, phenytoin, ethanol (chronic use), tobacco smoke
- Some drugs induce enzymes that metabolize them, a process called auto-induction (e.g. carbamazepine), and this results in development of tolerance

NON-CYTOCHROME P-450 ENZYMES INVOLVED IN OXIDATION PHASE I METABOLISM

Not all oxidation Phase I metabolism is catalyzed by cytochrome P-450 enzymes

Non-cytochrome P450 enzymes that are involved in oxidation Phase I metabolism include:

- Mono-amine oxidase
- Xanthine oxidase
- Alcohol dehydrogenase
- Aldehyde dehydrogenases

PHASE I REACTIONS: HYDROLYSIS

- These do not involve cytochrome P450 enzymes
- They occur in plasma and many tissues
- Hydrolytic enzymes include esterases, proteases/peptidases and amidases
- Esters (e.g. aspirin, procaine, succinylcholine) and amides (e.g. procainamide, lignocaine, indomethacin) undergo hydrolysis

PHASE II REACTIONS

- These are conjugation reactions
- The resulting conjugate is almost always pharmacologically inactive and less lipid soluble than its precursor and is excreted in urine or bile
- The groups most involved in conjugate formation are glucuronyl, sulphate, methyl, acetyl, glycyl and glutathione
- Many conjugation reactions occur in the liver, but many other tissues (e.g. kidney, lung) are also involved

PHASE II REACTIONS CONT'D

Examples of conjugation reactions:

- Glucuronidation - morphine, paracetamol, salicylates
- Glycine conjugation - salicylates
- Sulphate formation - oral contraceptive steroids, paracetamol
- Acetylation – isoniazid (the acetylation metabolite is hepatotoxic), phenelzine, dapsone
- Glutathione conjugation - paracetamol

LEARNING OUTCOMES

Students must be able to comprehend:

- Meaning of the term “Metabolism”
- Processes of drug metabolism
- Processes of Phase I and Phase II metabolism

QUIZ

1. Conjugation is:

- a) Process of drug reduction by special enzymes
- b) Process of drug oxidation by special oxidases
- c) Coupling of a drug with an endogenous substrate
- d) Solubilization in lipids

2. Which of the following processes proceeds in the second phase of biotransformation?

- a) Acetylation
- b) Reduction
- c) Oxidation
- d) Hydrolysis

3. Metabolic transformation and conjugation usually results in an increase of a substance biological activity:

- a) True
- b) False

4. Biotransformation of a medicinal substance results in:

- a) Faster urinary excretion
- b) Slower urinary excretion
- c) Easier distribution in organism
- d) Higher binding to membranes

END

Thanks for listening